

ColorSwapCore

RGB Color Processor with SPI Control and VGA Output
Chip Manual — v1.0

Tiny Tapeout Submission | SkyWater SKY130 PDK
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1. Project Overview

ColorSwapCore is a hardware RGB color transformation processor fabricated through the Tiny Tapeout open-source chip program on the SkyWater SKY130 process node. The chip receives 1-bit RGB pixel data via SPI from an ESP32, applies one of six selectable real-time color transformation modes, and outputs a standard VGA signal at 640x480 resolution and 60 Hz refresh rate. All arithmetic uses only shifts, additions, and bitwise logic — no multipliers — to fit within a single Tiny Tapeout tile.

2. Quick Start

1. Connect ESP32 SPI pins to ui_in[0..2] (CLK, MOSI, CS)
2. Connect ESP32 RGB GPIO pins to uio[0..2] (R, G, B)
3. Connect uo_out[0..4] to VGA monitor via 270-ohm resistors on RGB channels
4. Apply 25 MHz clock to clk pin; pull rst_n high after power-up
5. Chip starts in Mode 0 (original color) after reset
6. Send SPI byte 0x01 to switch to grayscale, 0x04 for negative, etc.

3. Pin Reference

3.1 Dedicated Inputs — ui_in[7:0]

Pin	Signal	Description
ui_in[0]	SPI_CLK	SPI clock input from ESP32
ui_in[1]	SPI_MOSI	SPI data input from ESP32 (MSB first)
ui_in[2]	SPI_CS	SPI chip select, active low
ui_in[7:3]	—	Reserved — tie to GND

3.2 Dedicated Outputs — uo_out[7:0]

Pin	Signal	Description
uo_out[0]	VGA_HSYNC	Horizontal sync, active low
uo_out[1]	VGA_VSYNC	Vertical sync, active low
uo_out[2]	VGA_R	Red channel, 1-bit output
uo_out[3]	VGA_G	Green channel, 1-bit output
uo_out[4]	VGA_B	Blue channel, 1-bit output
uo_out[7:5]	—	Reserved — driven low

3.3 Bidirectional Pins — uio[7:0] (all inputs, uio_oe = 0)

Pin	Signal	Description
uio[0]	R_IN	Red pixel input from ESP32, 1-bit
uio[1]	G_IN	Green pixel input from ESP32, 1-bit
uio[2]	B_IN	Blue pixel input from ESP32, 1-bit
uio[7:3]	—	Reserved

4. Color Mode Reference

Mode	SPI Byte	Name	Effect
0	0x00	Original	Passthrough — pixel unchanged
1	0x01	Grayscale	All channels = luma (OR of inputs)
2	0x02	Selective	Red pixels keep color; rest converted to grayscale
3	0x03	Retro Palette	4-color effect: black / cyan / magenta / white
4	0x04	Negative	Inverts all three channels
5	0x05	Thermal	Hot = red, Cold = blue, green suppressed

Note: Any SPI byte with bits[2:0] greater than 5 is ignored. The previous mode remains active.

5. SPI Protocol

The chip implements SPI Mode 0 (CPOL=0, CPHA=0). Frames are 8 bits, MSB first. Only bits[2:0] of the received byte are used as the mode selector. Maximum SPI clock frequency is 12.5 MHz (less than half the 25 MHz system clock, required by the 2-stage synchronizer).

Changing the Color Mode:

1. Pull CS low
2. Send 1 byte: bits[7:3] ignored, bits[2:0] = desired mode

3. Pull CS high — chip latches new mode immediately

Timing diagram:

```
CS: ---\_____/--- SCK: _/-\_-/-\_-/-\_-/-\_-/-\ MOSI:
[b7][b6][b5][b4][b3][b2:0=mode]
```

6. VGA Output Specification

Parameter	Value
Resolution	640 x 480 pixels
Refresh rate	60 Hz
Pixel clock	25.175 MHz
HSYNC polarity	Active LOW
VSYNC polarity	Active LOW
Color depth	1-bit per channel (8 possible colors)

Horizontal Timing

Region	Pixels
Active area	640
Front porch	16
Sync pulse	96
Back porch	48
Total	800

Vertical Timing

Region	Lines
Active area	480
Front porch	10
Sync pulse	2
Back porch	33
Total	525

7. Hardware Connection

VGA connector (DB-15):

DB-15 Pin	Signal	Connection
Pin 1 (Red)	VGA_R	uo_out[2] via 270-ohm resistor
Pin 2 (Green)	VGA_G	uo_out[3] via 270-ohm resistor
Pin 3 (Blue)	VGA_B	uo_out[4] via 270-ohm resistor
Pin 13 (HSYNC)	VGA_HSYNC	uo_out[0] direct
Pin 14 (VSYNC)	VGA_VSYNC	uo_out[1] direct
Pins 5,6,7,8,10	GND	System ground

ESP32 wiring:

ESP32 Signal	Chip Pin	Description
GPIO (SCK)	ui_in[0]	SPI clock
GPIO (MOSI)	ui_in[1]	SPI data
GPIO (CS)	ui_in[2]	SPI chip select
GPIO (R)	uio[0]	Red pixel bit
GPIO (G)	uio[1]	Green pixel bit
GPIO (B)	uio[2]	Blue pixel bit

8. Color Mode Logic Details

Mode 0 — Original

```
r_out = r_in; g_out = g_in; b_out = b_in;
```

Mode 1 — Grayscale

```
luma = r_in | g_in | b_in; r_out = g_out = b_out = luma;
```

Mode 2 — Selective (Red Isolation)

```
if (r_in) -> keep original color else -> r_out = g_out = b_out = luma
```

Mode 3 — Retro Palette

```
r_out = r_in | b_in; g_out = g_in & ~r_in; b_out = b_in | ~r_in; // Palette: black, cyan, magenta, white
```

Mode 4 — Negative

```
r_out = ~r_in; g_out = ~g_in; b_out = ~b_in;
```

Mode 5 — Thermal

```
r_out = r_in; // hot = red g_out = 0; // no green b_out = ~r_in & luma; // cold = blue
```

9. Reset Behavior

When `rst_n` is asserted low, all registers initialize to safe defaults:

- `mode_out` resets to 3'b000 (Mode 0 — Original color)
- `r_out`, `g_out`, `b_out` reset to 0
- VGA counters `h_count` and `v_count` reset to 0 (start of frame)
- SPI shift register and bit counter reset to 0
- SPI synchronizer registers reset to safe idle states

10. Electrical Specifications

Parameter	Value
Supply voltage	1.8 V (SKY130 core)
Technology	SkyWater SKY130 HD standard cells
Clock frequency	25 MHz
Reset type	Active low, synchronous
SPI clock max	12.5 MHz
RGB output type	1-bit per channel CMOS
Recommended series resistor	270 ohm per RGB channel
Tile size	Single tile (~167 x 108 micrometers)

11. Design Constraints and Limitations

Constraint	Detail
1-bit per channel	Only 8 possible colors (2^3). No gradients or intermediate shades.
No framebuffer	All pixel data must be streamed from ESP32 in real time, synchronized with VGA scan.
SPI clock limit	Must be less than 12.5 MHz due to 2-stage synchronizer requirement.
No multipliers	All color math uses bitwise operations only to minimize silicon area.
Single tile	Design fits within one Tiny Tapeout tile (~167x108 um on SKY130).